

SUDHA LAKSHMI R.

Computer Science Undergraduate | Python · Java · Data Structures & Algorithms · Machine Learning & LLMs

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PROFESSIONAL SUMMARY

Computer Science undergraduate with a strong foundation in Data Structures & Algorithms and Object-Oriented Programming (encapsulation, inheritance, polymorphism, abstraction). Hands-on experience building AI-driven applications, including a full-stack platform powered by Large Language Models (Llama 3) and real-time machine learning inference. Curious and fast-learning, with practical exposure to Python through applied data-workflow projects, and a strong interest in Agentic AI, LLMs, and Knowledge Graphs. Comfortable working in fast-paced, hands-on environments and eager to contribute to production-grade AI systems.

SKILLS

Programming: Python, Java, C

AI / ML: LLM-integrated application development (Llama 3 via Groq), client-side ML inference (WebAssembly), real-time pose & gaze detection (Google MediaPipe)

Core CS: Data Structures & Algorithms, Object-Oriented Programming, SQL

Web / Backend: React, Node.js, HTML/CSS, WebSockets, PostgreSQL (Supabase)

Tools: VS Code

Soft Skills: Structured problem-solving, teamwork, communication, fast-paced/startup adaptability

PROJECTS

Echo — AI-Powered Technical Interview Platform

Next.js, React, PostgreSQL (Supabase), Groq (Llama 3), Google MediaPipe

- Architected a full-stack AI interview platform using Llama 3 (LLM) for dynamic technical assessments and automated scoring, enabling personalized, AI-driven candidate evaluation at scale.
- Engineered client-side machine learning via WebAssembly to enforce real-time anticheat monitoring, applying pose and gaze detection models to preserve assessment integrity.

CodeCollab — Real-Time Collaborative Code Editor

React, Node.js, WebSockets, Monaco Editor

- Built a browser-based collaborative coding environment supporting real-time multi-user editing with conflict resolution using the Operational Transformation (OT) algorithm.
- Integrated syntax highlighting for 10+ languages and WebSocket-based live session sharing, enabling seamless remote pair-programming.

ACHIEVEMENTS

Best Paper Award — Kamaraj Innovate '26 | Conference Publication

- Published and presented "PosePerfect: Real-Time Posture Recognition Using MediaPipe and Machine Learning — A Quality-Driven Approach Integrating 7 QC Tools, FMEA, and Six Sigma DMAIC" at Kamaraj Innovate '26.
- Awarded Best Paper for applying quality-engineering methodologies (7 QC Tools, FMEA, Six Sigma DMAIC) to a real-time ML posture-recognition system built on Google MediaPipe.

INTERNSHIPS

Software Development Job Simulation | JP Morgan

- Designed and implemented a data feed interface using Python and Perspective to visualize real-time stock price data for traders in a financial software workflow.
- Applied software engineering best practices — version control, debugging, and test-ready code — aligned with JP Morgan's technical standards.

Frontend Development Job Simulation | Skyscanner

- Developed a responsive travel search UI component using React and the Backpack design system, building accessible, mobile-first frontend features used in production at scale.
- Strengthened component-driven architecture skills, including writing unit tests and collaborating against design guidelines to deliver pixel-perfect UI.

EDUCATION

Bachelor of Technology in Computer Science and Engineering | *SRM Institute of Science and Technology* 2023 – 2027

LANGUAGES KNOWN

Tamil | English | Hindi | Malayalam